Overall Verification Results

Executive summary

This document consolidates the complete verification record for Task A and Task B, including the blocker status, environment record, root-cause diagnosis, and computational checks.

Hash integrity

The source script hash matches the required value. The supplied results.json hash does not match the canonical required value, which blocks Task B.

Root cause

The function A_prime fails to treat manufactured_a as a recognized branch and falls through to a zero return path.

Task A findings

Recommendation is BLOCK. The original archive is restricted, primary figure arrays are unavailable, and the main protocol gates are not frozen.

Task B status

Blocked because the controlled output artifact hash mismatch shuts down any valid original-grid validation.

Conclusion

The bug is proven; the verification gate is also proven. The proper status is BLOCKED with evidence, not a false pass.

Key evidence table

Item	Result
oscillator_verification.py hash	CONFIRMED
results.json hash	MISMATCH
A_prime(manufactured_a, 0.0)	0.0
manufactured_ap(0.0)	0.48121182505960347
Expected derivative	0.48121182505960347
Task A recommendation	BLOCK
Task B verdict	BLOCKED

Environment

- OS: Windows 11 AMD64 - Python: 3.13.7 - NumPy: 2.4.4 - SciPy: 1.17.1 - SymPy: 1.14.0 - Matplotlib: 3.10.9 - ReportLab: 4.4.4 - Random seed: 0

Hash integrity

The original oscillator verification script hash matches the required value: - oscillator_verification.py: 8d7fd2612e37ab3a30a08ba4829295c0c7415819486e3a96adcc1f78e8fca5ea The output artifact hash does not match the required canonical value: - expected: 2074913e8d2e87048d9a9e419257586cfc02f7005789d90ba9ecf94e8efc99cf - actual: aeac34a0e92fb2e81051c873e94fc39a823be86f435cd42c0ebc578b8675952d This mismatch is the principal blocker for Task B because the task requires the original outputs and forbids modifying them after the verification gate.

Root-cause finding

The original oscillator script contains a dispatch bug in A_prime. The function recognizes unit_A, constant_A, and eta_A but does not recognize manufactured_a. The final branch returns zeros when none of those identities match, so the manufactured forcing branch quietly disappears. Evidence from the running code: - A_prime(manufactured_a, 0.0) = 0.0 - manufactured_ap(0.0) = 0.48121182505960347 - expected derivative = 0.48121182505960347 This is a direct runtime proof that the dispatch is wrong and that the manufactured branch should return manufactured_ap(t_arr).

Task A findings

Task A was a review-only protocol assessment for DTL-ST-REPRO-001 and the recommendation is BLOCK. Key findings: 1. The original code archive is restricted and the protocol package explicitly says the original replay is blocked. 2. The primary figure arrays and source-data idiosyncrasies remain unavailable; Figure 1c and Figure 4 arrays are not available. 3. The protocol's gate matrix shows that the primary gates are not frozen and N_traj = 100 is too small for rare-event claims without exact rate bounds.

Task B status

Task B remains BLOCKED despite the clear root cause. The required hash mismatch on the original results artifact prevents valid completion of the original-grid validation. The rule forbids modifying the original outputs, so no false green result can be produced.

Conclusion

Contained in overall summary above.

Direct evidence summary

A_prime(manufactured_a, 0.0) = 0.0

manufactured_ap(0.0) = 0.48121182505960347

expected derivative = 0.48121182505960347

Therefore the branch selection is incorrect and the zero-return fallback is the cause of the observed defect.

Compute estimate checks

Expression	Value
L = 100, Ntraj = 100, Np = 10^4	baseline
100^2 * 100 * 10^4 * 20	2.0e11
100^2 * 100 * 10^4 * 40	4.0e11
τ_osc = 0.001, τ_cav/τ_osc = 20	γ = 100
τ_osc = 0.001, τ_cav/τ_osc = 200	γ = 10

Final determination: verified root cause, verified blocker, TASK B remains BLOCKED.